Practice 1: Fraction meaning

Name: ____________________ Date: __________

For each problem, show your model or reasoning.

1. A rectangle is split into 8 equal parts. 3 are shaded. Write the fraction shaded.

Work space

2. In $5/7$, identify the numerator and denominator. Explain what each tells you.

Work space

3. Draw a bar model for $4/6$. Are the parts equal? Explain.

Work space

Practice 2: Fractions of sets

Name: ____________________ Date: __________

1. What is $3/5$ of 20 counters? Draw groups to show it.

Work space

2. 7 of 12 marbles are blue. Write the fraction that are blue and the fraction that are not blue.

Work space

3. A class has 24 students. $1/4$ bring a lunch from home. How many students is that?

Work space

Practice 3: Unit fractions

Name: ____________________ Date: __________

1. Write the unit fraction for one of 9 equal parts.

Work space

2. How many $1/8$ pieces make one whole? Explain with a strip.

Work space

3. Which is larger, $1/3$ or $1/6$? Use a drawing or words to justify.

Work space

Practice 4: Improper fractions and mixed numbers

Name: ____________________ Date: __________

1. Change $7/4$ to a mixed number.

Work space

2. Change $2\frac{3}{5}$ to an improper fraction.

Work space

3. Draw a model for $1\frac{1}{2}$ and label the improper fraction.

Work space

Practice 5: Number lines

Name: ____________________ Date: __________

1. Divide 0 to 1 into 6 equal intervals and mark $4/6$.

Work space

2. Which fraction is at the third tick after 0 when a number line has eighths?

Work space

3. Mark $5/4$ on a number line. Write it as a mixed number.

Work space

Practice 6: Equivalent fractions by multiplying

Name: ____________________ Date: __________

1. Complete $2/3 = ?/12$.

Work space

2. Complete $5/8 = 15/?$.

Work space

3. Write two fractions equivalent to $3/7$.

Work space

Practice 7: Equivalent fractions by dividing

Name: ____________________ Date: __________

1. Simplify $12/18$.

Work space

2. Simplify $20/25$.

Work space

3. Is $18/24$ equivalent to $3/4$? Show how you know.

Work space

Practice 8: Visual equivalence

Name: ____________________ Date: __________

1. Explain why $1/2$ and $4/8$ name the same amount.

Work space

2. A strip has 3 of 6 equal parts shaded. Write two equivalent fractions.

Work space

3. True or false: $2/5 = 4/10$. Prove your answer with a model or calculation.

Work space

Practice 9: Missing numerators and denominators

Name: ____________________ Date: __________

1. $3/4 = ?/20$.

Work space

2. $7/9 = 21/?$.

Work space

3. $4/11 = ?/33$.

Work space

Practice 10: Missing denominators

Name: ____________________ Date: __________

1. $6/10 = 3/?$.

Work space

2. $8/12 = 2/?$.

Work space

3. $15/35 = 3/?$.

Work space

Practice 11: Simplest form

Name: ____________________ Date: __________

1. Reduce $16/20$ to simplest form.

Work space

2. Reduce $21/28$ to simplest form.

Work space

3. Is $9/13$ already in simplest form? Explain.

Work space

Practice 12: Equivalent fractions in context

Name: ____________________ Date: __________

1. Mia ate $2/6$ of a pizza. Ben ate $1/3$ of the same pizza. Did they eat the same amount? Explain.

Work space

2. A recipe uses $3/4$ cup. How many eighths of a cup is that?

Work space

3. A trail is $6/10$ mile completed. Give the simplest equivalent fraction.

Work space

Practice 13: Compare using equivalence

Name: ____________________ Date: __________

1. Compare $2/3$ and $8/12$ using $<$, $>$, or $=$.

Work space

2. Compare $5/6$ and $15/18$. Explain.

Work space

3. Which is greater: $3/5$ or $9/15$? Show an equivalent fraction.

Work space

Practice 14: Fractions on models

Name: ____________________ Date: __________

1. Draw two equal-size rectangles to show $2/3 = 6/9$. Show your two rectangles clearly.

Work space

2. Draw and shade a model showing $3/4 = 9/12$. Label the parts you shade.

Work space

3. Draw a set model showing $2/5 = 6/15$. Make each group contain the same number of objects.

Work space

Practice 15: Explain and justify

Name: ____________________ Date: __________

1. Explain why multiplying both numerator and denominator by 3 keeps a fraction's value unchanged.

Work space

2. A student says $4/8$ is greater than $1/2$ because 4 is greater than 1. Explain the error.

Work space

3. Give a fraction equivalent to $5/6$ with denominator 24 and explain your steps.

Work space

Practice 16: Mixed and improper equivalence

Name: ____________________ Date: __________

1. Write $9/4$ as a mixed number and give an equivalent improper fraction for your mixed number.

Work space

2. Complete $1\frac{2}{3} = ?/3$.

Work space

3. Are $2\frac{1}{2}$ and $5/2$ equivalent? Show why.

Work space

Practice 17: Find the whole

Name: ____________________ Date: __________

1. If 4 counters represent $2/3$ of a collection, how many counters are in the whole collection? Explain.

Work space

2. 6 red tiles are $3/5$ of all the tiles. How many tiles are there?

Work space

3. A ribbon's $5/8$ is 15 cm. What is the full length?

Work space

Practice 18: Error analysis

Name: ____________________ Date: __________

1. Jordan changes $3/5$ to $6/10$ by adding 3 to the numerator and 5 to the denominator. Is this reasoning valid? Explain.

Work space

2. A student simplifies $12/16$ to $6/8$ and stops. Is it simplest form? Finish the work.

Work space

3. Find and correct the mistake: $2/7 = 2/14$.

Work space

Practice 19: Review

Name: ____________________ Date: __________

1. Simplify $24/32$, then write an equivalent fraction with denominator 20.

Work space

2. A garden has 18 of 30 plants flowering. Write the fraction, simplify it, and name an equivalent fraction with denominator 5.

Work space

3. Place $3/2$, $4/4$, and $3/4$ in order from least to greatest. Explain.

Work space

Practice 20: Final challenge

Name: ____________________ Date: __________

1. A 24-slice tray has 18 slices left. Write the fraction left in simplest form and an equivalent fraction with denominator 8.

Work space

2. Create three different fractions equivalent to $4/5$. Show the multiplier used each time.

Work space

3. Write a short rule for making equivalent fractions, then use it to show $7/10 = ?/40$.

Work space